# An Exact Minimum for Weighted Closed-Loop Phase Locking

**Abstract**

For a prescribed closed loop, positive weight, geometric phase target, connection, and winding number, we determine the exact minimum of a weighted quadratic phase-mismatch functional. A sharp Cauchy–Schwarz inequality, together with an explicit admissible phase, gives the minimum and a unique minimizing covariant phase gradient. An exact remainder identity establishes strict minimality modulo constant-phase symmetry. Comparison between different winding sectors is conditional on the locking functional accounting for all sector-dependent contributions and does not itself provide a transition mechanism.

## 1 Setting and admissible phases

Let $\Gamma$ be an oriented closed loop of total arclength $S > 0$, with coordinate $s \in [0, S]$ and its endpoints identified. Fix integers $m, n \in \mathbb{Z}$. Let $w : [0, S] \to \mathbb{R}$ be measurable, and suppose there are constants $0 < w_- \le w_+ < \infty$ such that

$$w_- \le w(s) \le w_+ \qquad \text{for almost every } s. \tag{1}$$

Let $\Omega, \sigma \in L^2(0, S; \mathbb{R})$ be fixed. We use a fixed framing and endpoint convention in which an admissible real phase lift $\chi \in H^1(0, S; \mathbb{R})$ satisfies

$$\chi(S) - \chi(0) = 2\pi n. \tag{2}$$

The phase $e^{i\chi}$ is therefore periodic. Define

$$q(s) := \chi'(s) - m\Omega(s), \qquad \mathcal{E}[\chi] := \int_0^S w(s)\big(q(s) - \sigma(s)\big)^2 \, \mathrm{d}s. \tag{3}$$

Introduce the fixed quantities

$$\Phi := \int_0^S \Omega(s) \, \mathrm{d}s, \qquad J := \int_0^S \sigma(s) \, \mathrm{d}s, \qquad I := \int_0^S \frac{\mathrm{d}s}{w(s)}. \tag{4}$$

By (1), $0 < I < \infty$. The admissible space in winding sector $n$ is

$$\mathcal{A}_n := \{\chi \in H^1(0, S; \mathbb{R}) : \chi(S) - \chi(0) = 2\pi n\}. \tag{5}$$

All geometric data, the framing, $m$, and $n$ are held fixed during minimization.

## 2 Exact minimum and sharp remainder

**Theorem 1** (Exact weighted phase-locking minimum)**.** *Under the assumptions above, set*

$$\Delta_n := 2\pi n - m\Phi - J. \tag{6}$$

*Then*

$$\boxed{\min_{\chi\in\mathcal{A}_n} \mathcal{E}[\chi] = \frac{\Delta_n^2}{I} = \frac{(2\pi n - m\Phi - J)^2}{\displaystyle\int_0^S w^{-1}\,\mathrm{d}s}.} \tag{7}$$

*The minimizing covariant phase gradient is, uniquely up to equality almost everywhere,*

$$\boxed{q_*(s) = \sigma(s) + \frac{\Delta_n}{I\,w(s)}.} \tag{8}$$

*An admissible minimizing phase is*

$$\boxed{\chi_*(s) = \chi_0 + \int_0^s \left[m\Omega(t) + \sigma(t) + \frac{\Delta_n}{I\,w(t)}\right]\,\mathrm{d}t,} \tag{9}$$

*where $\chi_0 \in \mathbb{R}$ is arbitrary. Every minimizing phase differs from $\chi_*$ by an additive constant. Furthermore, every $\chi \in \mathcal{A}_n$ satisfies the exact identity*

$$\boxed{\mathcal{E}[\chi] = \frac{\Delta_n^2}{I} + \int_0^S w(s)|q(s) - q_*(s)|^2\,\mathrm{d}s.} \tag{10}$$

*Proof.* Integrating the definition of $q$ and using the winding constraint gives

$$\int_0^S q\,\mathrm{d}s = \chi(S) - \chi(0) - m\int_0^S \Omega\,\mathrm{d}s = 2\pi n - m\Phi. \tag{11}$$

Hence for the mismatch $f := q - \sigma$,

$$\int_0^S f\,\mathrm{d}s = 2\pi n - m\Phi - J = \Delta_n. \tag{12}$$

By the Cauchy–Schwarz inequality,

$$\Delta_n^2 = \left(\int_0^S \sqrt{w(s)} f(s) \frac{1}{\sqrt{w(s)}}\,\mathrm{d}s\right)^2 \tag{13}$$

$$\leq \left(\int_0^S w(s) f(s)^2\,\mathrm{d}s\right)\left(\int_0^S \frac{\mathrm{d}s}{w(s)}\right) = I\mathcal{E}[\chi]. \tag{14}$$

Because $I > 0$, this yields

$$\mathcal{E}[\chi] \geq \frac{\Delta_n^2}{I}. \tag{15}$$

Cauchy–Schwarz is an equality precisely when $\sqrt{w}\,f = C/\sqrt{w}$ almost everywhere for some constant $C$. Thus $f = C/w$, and (12) implies

$$\Delta_n = C\int_0^S w^{-1}\,\mathrm{d}s = CI, \qquad C = \frac{\Delta_n}{I}. \tag{16}$$

The only possible minimizing covariant gradient is consequently (8).

To prove that the lower bound is attained, define $\chi_*$ by (9). The stated regularity assumptions ensure $\chi_* \in H^1(0,S)$, and

$$\chi_*(S) - \chi_*(0) = m\Phi + J + \frac{\Delta_n}{I}\int_0^S \frac{\mathrm{d}s}{w(s)} \tag{17}$$

$$= m\Phi + J + \Delta_n = 2\pi n. \tag{18}$$

Thus $\chi_* \in \mathcal{A}_n$, and $q_* = \chi_*' - m\Omega$. Its energy is

$$\mathcal{E}[\chi_*] = \int_0^S w\left(\frac{\Delta_n}{Iw}\right)^2 \mathrm{d}s = \frac{\Delta_n^2}{I^2}\int_0^S w^{-1}\,\mathrm{d}s = \frac{\Delta_n^2}{I}. \tag{19}$$

This proves the exact minimum.

For the remainder identity, let $h := q - q_*$. Since $\chi$ and $\chi_*$ have the same winding number,

$$\int_0^S h\,\mathrm{d}s = \int_0^S (\chi' - \chi_*')\,\mathrm{d}s = 0. \tag{20}$$

Also, $w(q_* - \sigma) = \Delta_n/I$. Expanding the square yields

$$\mathcal{E}[\chi] = \int_0^S w\big[(q_* - \sigma) + h\big]^2\,\mathrm{d}s \tag{21}$$

$$= \mathcal{E}[\chi_*] + \frac{2\Delta_n}{I}\int_0^S h\,\mathrm{d}s + \int_0^S wh^2\,\mathrm{d}s \tag{22}$$

$$= \frac{\Delta_n^2}{I} + \int_0^S w|q - q_*|^2\,\mathrm{d}s. \tag{23}$$

Positivity of $w$ shows that equality holds only for $q = q_*$ almost everywhere. In that case $(\chi - \chi_*)' = 0$ almost everywhere, so $\chi - \chi_*$ is constant. This proves the asserted uniqueness. □

## 3 Consequences

**Corollary 2** (Criterion for an exact lock)**.** *There exists an admissible phase with $q = \sigma$ almost everywhere if and only if*

$$\boxed{2\pi n = m\Phi + J.} \tag{24}$$

*Equivalently,* $\min_{\mathcal{A}_n} \mathcal{E} = 0$ *if and only if* $\Delta_n = 0$.

*Proof.* The conclusion follows from (7), or directly by integrating $q = \sigma$ in (11). □

**Corollary 3** (Strict phase-sector variational stability)**.** *Let $\eta \in H^1(0,S;\mathbb{R})$ have periodic endpoint values $\eta(S) = \eta(0)$. Then for every $\varepsilon \in \mathbb{R}$,*

$$\mathcal{E}[\chi_* + \varepsilon\eta] = \mathcal{E}[\chi_*] + \varepsilon^2\int_0^S w(s)|\eta'(s)|^2\,\mathrm{d}s. \tag{25}$$

*In particular,*

$$\frac{\mathrm{d}^2}{\mathrm{d}\varepsilon^2}\mathcal{E}[\chi_* + \varepsilon\eta]\Big|_{\varepsilon=0} = 2\int_0^S w|\eta'|^2\,\mathrm{d}s \geq 0, \tag{26}$$

*with equality if and only if $\eta$ is constant almost everywhere.*

*Proof.* The perturbation preserves the winding constraint, and its covariant gradient differs from $q_*$ by $\varepsilon\eta'$. Apply (10). □

**Remark 4** (Meaning of stability)**.** *Corollary 3 is a statement about a fixed-geometry, phase-only variational functional. It does not assert dynamical, orbital, or spectral stability for a full complex-field evolution. In particular, it does not allow the winding number to change during the perturbation.*

**Proposition 5** (Conditional comparison of winding sectors)**.** *Suppose the geometric data $\Gamma, w, \sigma, \Omega$ and integer $m$ are fixed, all integer winding sectors are admissible, and the complete sector-dependent energy is $\mathcal{E}$ plus a term independent of $n$. If the system is allowed to compare the global minimum energies of these sectors, define*

$$\zeta := \frac{m\Phi + J}{2\pi}. \tag{27}$$

*Then the integer sectors of least energy are exactly*

$$\boxed{\mathcal{N}_{\min} = \arg\min_{n\in\mathbb{Z}} |n - \zeta|.} \tag{28}$$

*There is one minimizing integer unless $\zeta \in \mathbb{Z} + \frac{1}{2}$, when two adjacent integers are tied.*

*Proof.* Theorem 1 gives

$$\mathcal{E}_{\min}(n) = \frac{(2\pi n - m\Phi - J)^2}{I} = \frac{4\pi^2}{I}(n - \zeta)^2. \tag{29}$$

The prefactor is positive and independent of $n$, so minimizing over integers is equivalent to selecting the integer(s) nearest to $\zeta$. □

**Remark 6** (Limits of inter-sector selection)**.** *Proposition 5 compares minimum costs; it does not establish that the physical dynamics can change $n$. With a continuous nonvanishing complex field on a fixed closed contour, winding is topologically conserved under continuous evolution. A change of winding requires additional conditions or dynamics, for example a field zero (phase slip), altered boundary data, or a change of the contour topology. Moreover, omitted $n$-dependent energies would change the sector comparison.*

**Remark 7** (Frame conventions)**.** *The constraint $\int q\,\mathrm{d}s = 2\pi n - m\Phi$ assumes the stated phase-lift and framing convention. If the normal frame has a nontrivial seam identification, its transition phase must be incorporated consistently into the endpoint condition and holonomy. The proof then applies with the correspondingly adjusted, gauge-consistent total closure mismatch.*

## 4 Uniform-weight example

If $w(s) \equiv w_0 > 0$, $\Omega \equiv 0$, and $\sigma(s) \equiv \sigma_0$, then $I = S/w_0$, $J = S\sigma_0$, and

$$\mathcal{E}_{\min}(n) = \frac{w_0}{S}(2\pi n - S\sigma_0)^2, \qquad q_*(s) = \frac{2\pi n}{S}. \tag{30}$$

Thus the optimal phase gradient is uniform, but it need not coincide with the preferred local rate $\sigma_0$. Exact locking occurs precisely when $S\sigma_0 = 2\pi n$.

## Scope

The theorem is an exact result for the specified weighted phase-only functional, closed-loop geometry, winding sector, and framing. Its proof uses the sharp weighted Cauchy–Schwarz inequality; it does not provide an independent physical derivation of the locking energy, a mechanism for transitions between winding sectors, or a prediction for geometry or spectral frequencies.
**Supporting analysis:** https://github.com/rickyjreyes/geometry_of_resonance/blob/main/docs/WINDING_SELECTION_ANALYSIS.md